

Lab: Random Variables and Probability Distributions

Objective

Learn to represent an agent's beliefs and preferences using **probability distributions**, and practice computing basic probabilities that underpin the Active Inference framework.

Prerequisites

- Completed Math Foundations: Systems module
- Basic arithmetic with fractions and decimals

Part 1: Discrete Probability Distributions

Goal: Build and analyze probability distributions over agent states.

A student agent has beliefs about tomorrow's weather. Define a random variable $X = \{\text{sunny, cloudy, rainy}\}$ with probabilities:

1. Assign probabilities: $P(\text{sunny}) = 0.5$, $P(\text{cloudy}) = 0.3$, $P(\text{rainy}) = 0.2$.
2. Verify they sum to 1.
3. What is the most likely state (the mode)?
4. Draw a bar chart of this distribution.

Write your response here

Part 2: Joint and Marginal Probabilities

Goal: Work with two random variables simultaneously.

Let $X = \text{weather} = \{\text{sunny, rainy}\}$ and $Y = \text{mood} = \{\text{happy, sad}\}$. The joint distribution is:

	happy	sad
sunny	0.4	0.1
rainy	0.1	0.4

1. Compute $P(\text{sunny})$ and $P(\text{rainy})$ by summing rows.
2. Compute $P(\text{happy})$ and $P(\text{sad})$ by summing columns.
3. Are X and Y independent? Check: $P(\text{sunny, happy}) = P(\text{sunny}) * P(\text{happy})$?

Write your response here

Part 3: Expected Value

Goal: Compute expected values as a measure of what an agent "expects."

A game pays \$10 if you roll a 6 on a fair die, \$0 otherwise.

1. $P(\text{win}) = ?$, $P(\text{lose}) = ?$
2. $E[\text{payout}] = P(\text{win}) * \$10 + P(\text{lose}) * \$0 = ?$
3. Would you pay \$2 to play? Why?
4. How does expected value connect to policy evaluation in Active Inference?

Write your response here

Part 4: Preferred Distributions

Goal: Represent agent preferences as a probability distribution.

Define a preferred distribution $P^*(\text{grade}) = \{A: 0.7, B: 0.2, C: 0.1\}$ and actual $P(\text{grade}) = \{A: 0.3, B: 0.4, C: 0.3\}$.

- 1. Where are the biggest mismatches?
- 2. What actions could reduce the gap between P and P*?
- 3. In Active Inference, this gap is related to free energy. The agent acts to make P closer to P*.

Write your response here

Summary Table

Math Concept	Symbol	Definition
Random variable	X	Variable whose value is determined by chance
Probability distribution	P(X)	Assigns probabilities summing to 1
Joint distribution	P(X, Y)	Probability over two variables
Expected value	E[X]	Probability-weighted average
Preferred distribution	P*(X)	Target distribution the agent acts to achieve